

Bloch Sphere:

$$\vec{S}(t) = \begin{pmatrix} \langle \hat{S}_x \rangle_{\psi(t)} \\ \langle \hat{S}_y \rangle_{\psi(t)} \\ \langle \hat{S}_z \rangle_{\psi(t)} \end{pmatrix}$$

$$\psi(t) = \frac{e^{-i\frac{\omega_0 t}{2}} |+\rangle + e^{+i\frac{\omega_0 t}{2}} |-\rangle}{\sqrt{2}}$$

$$\vec{S}(t) = \frac{\hbar}{2} \begin{pmatrix} \cos(\omega_0 t) \\ \sin(\omega_0 t) \\ 0 \end{pmatrix}$$

$$\langle \vec{S}^2 \rangle = \langle S_x^2 \rangle + \langle S_y^2 \rangle + \langle S_z^2 \rangle = \frac{3\hbar^2}{4}$$

Because $\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = 1$

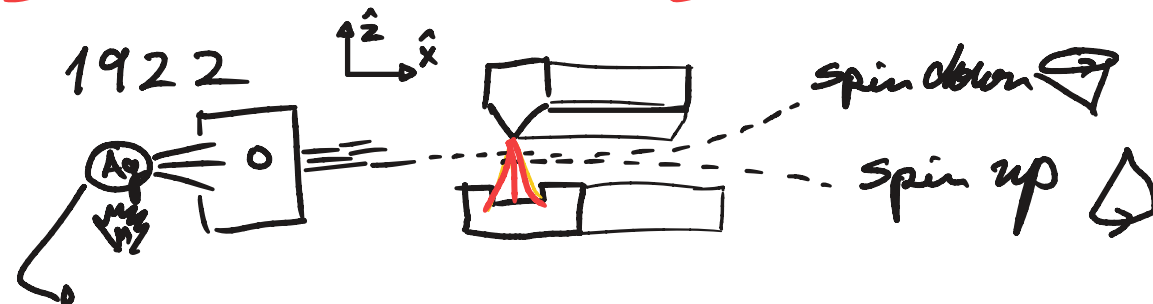
Also

$$|\vec{S}|^2 = \langle S_x \rangle^2 + \langle S_y \rangle^2 + \langle S_z \rangle^2 = \hbar^2/4$$

$$\sum_i (\Delta S_i)^2 = \sum_i \langle \hat{S}_i^2 \rangle - \langle S_i \rangle^2 = \frac{3\hbar^2}{4} - \frac{\hbar^2}{4} = \frac{\hbar^2}{2}$$

Not true for mixed states which have $|\vec{S}|^2 < \hbar^2/4$

Stern-Gerlach



$Z = 47 \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^1$
 $N \sim 61$
 $A \sim 108$
 $\downarrow \uparrow \quad \downarrow \uparrow \quad \downarrow \uparrow$

$\mu \sim \frac{e\hbar}{2m}$; $F = -\vec{\nabla} U = \nabla(\vec{\mu} \cdot \vec{B})$

$\vec{F} = \vec{\nabla}(\vec{\mu} \cdot \vec{B}) \approx \mu_z \frac{\partial B}{\partial z}$

$\vec{\mu} = \gamma \vec{S} = g \frac{q}{4m} \hbar \vec{\sigma} = -\frac{e\hbar}{2m} \vec{\sigma}$

$\vec{F} = -\frac{e\hbar}{2m} \vec{\sigma} (\vec{\nabla} \cdot \vec{B}) = -\frac{e\hbar}{2m} \sigma_z \frac{\partial B}{\partial z} \hat{z}$

so: $\vec{F} | \uparrow \rangle = -\frac{e\hbar}{2m} \frac{\partial B}{\partial z} \hat{z}$

$\vec{F} | \downarrow \rangle = +\frac{e\hbar}{2m} \frac{\partial B}{\partial z} \hat{z}$

if $g = +2$ if $q = -e$; $g = -2$ if $q = +e$.

How to describe this exp?

Projectors: $|+\rangle$ spin up
 $|-\rangle$ spin down

$$P_+ = |+\rangle\langle+| \quad ; \quad P_- = |-\rangle\langle-|$$

- Hermitian: $P_+^\dagger = (|+\rangle\langle+|)^\dagger = |+\rangle\langle+| = P_+$
- Idempotent: $P_+^2 = (|+\rangle\langle+|)(|+\rangle\langle+|) = P_+$
- Orthogonal: $P_+ P_- = |+\rangle\langle+|-\rangle\langle-| = 0$
- Complete: $P_+ + P_- = \mathbb{I}$

Take $|\psi\rangle = a|+\rangle + b|-\rangle$ with $|a|^2 + |b|^2 = 1$
then $P_+|\psi\rangle = a|+\rangle \rightarrow \langle\psi|P_+|\psi\rangle = |a|^2$
 $\langle\psi|P_-|\psi\rangle = |b|^2$

But $\|P_+|\psi\rangle\| = |a| \leq 1$
 $\Rightarrow$ Say new state is:
 $\frac{P_+|\psi\rangle}{\|P_+|\psi\rangle\|} = \frac{a}{|a|}|+\rangle = e^{i\gamma}|+\rangle$

Any operator $P^2 = P$ must have eigenvalues 0 or 1

$$P_+|\psi\rangle = \lambda|\psi\rangle \\ \Rightarrow P_+^2|\psi\rangle = \lambda^2|\psi\rangle \Rightarrow \lambda^2 = \lambda \\ +1 \text{ or } 0$$

$$\hat{n} = (\sin\beta \cos\alpha; \sin\beta \sin\alpha, \cos\beta) \\ \Rightarrow \hat{S}_n = \hat{n} \cdot \vec{S} = n_x \hat{S}_x + n_y \hat{S}_y + n_z \hat{S}_z \\ \hat{S}_n = \frac{\hbar}{2} \left[c_\beta P_+ + e^{i\alpha} s_\beta |+\rangle\langle-| + e^{i\alpha} s_\beta |-\rangle\langle+| - c_\beta P_- \right]$$

n.b. traceless.

$$\text{Eigenvalues: } \hat{S}_n^2 = (\vec{n} \cdot \vec{S})^2 = \frac{\hbar^2}{4} (\vec{n} \cdot \vec{\sigma})^2 \\ = \frac{\hbar^2}{4} \left(\sum_i n_i \sigma_i \right)^2 = \frac{\hbar^2}{4} \left(\sum_{i,j} n_i n_j \sigma_i \sigma_j \right) \\ = \frac{\hbar^2}{4} \left(\sum_{i,j} n_i n_j (\delta_{ij} \mathbb{I} + i \epsilon_{ijk} \sigma_k) \right)^2 = \frac{\hbar^2}{4} |n|^2 = \frac{\hbar^2}{4}$$

$\hat{S}_n$ has eigenvalues $\pm \hbar/2$ no matter what $\hat{n}$ is.

$$\vec{n} \cdot \vec{S} |n\rangle = \lambda |n\rangle \Rightarrow \frac{\hbar^2}{4} |n\rangle = \lambda^2 |n\rangle$$

$$\lambda = \pm \hbar/4; \quad \text{Tr}(\vec{S}_n) = \lambda_1 + \lambda_2 = 0.$$

$\langle \hat{S}_n \rangle$ cannot depend on $\vec{n}$.

Eigenvectors

$$|S_{\hat{n}}; +\rangle = \cos \frac{\beta}{2} |+\rangle + e^{i\alpha} \sin \frac{\beta}{2} |-\rangle$$

$$|S_{\hat{n}}; -\rangle = -e^{i\alpha} \sin \left(\frac{\beta}{2}\right) |+\rangle + \cos \frac{\beta}{2} |-\rangle$$

$$\neq S_{\hat{n}} |+\rangle \text{ or } S_{\hat{n}} |-\rangle$$

Stern-Gerlach Filter in direction $\hat{n}$:

$$\hat{P}_{\hat{n}; \pm} = |S_{\hat{n}}; \pm\rangle \langle S_{\hat{n}}; \pm| \rightarrow \underline{SG \hat{n}}(+)$$

BTW $\langle \vec{S}_n \rangle = \frac{\hbar}{2} \hat{n}$

Block vector points in direction of magnet

The fraction/probability of detecting state $|S_{\hat{n}}; +\rangle$ after the following chain $SG \hat{z}(+) \rightarrow SG \hat{n} \rightarrow \text{measure}$ is given by

Describes $SG \hat{z}(+)$ filter

$$P_+ = \underbrace{\langle S_{\hat{n}}; + | P_+ | S_{\hat{n}}; + \rangle}_{\text{Expectation Value of } P_+} = \underbrace{|\langle S_{\hat{n}}; + | + \rangle|^2}_{\text{How much } S_{\hat{n}}; + \text{ is in the previously prepared } |+\rangle \text{ state?}} = \cos^2 \frac{\beta}{2}$$

$$P_- = |\langle - | S_{\hat{n}}; - \rangle|^2 = \sin^2 \frac{\beta}{2}$$

for $\beta = \pi$

$$\text{so } \begin{cases} P_- = 1 \\ P_+ = 0 \end{cases}$$

$$SG \hat{z}(+) \xrightarrow{\hat{n} = -\hat{z}} SG(-\hat{z}) \rightarrow \text{measure}$$

$$|S_{-\hat{z}}; -\rangle = |S_{\hat{z}}; +\rangle$$

for $\beta = \pi/2$

$$SG \hat{z}(+) \xrightarrow{\hat{n} = \hat{x}} SG \hat{x} \rightarrow \text{measure}$$

$$P(\hat{n} \rightarrow +) = 1/2$$

$$P(\hat{n} \rightarrow -) = 1/2$$

50/50 split

Again...

a) $\underbrace{SG \hat{z}(+) \rightarrow SG \hat{z}}_{|+\rangle} \rightarrow \text{measure}$
 $\langle + | P_+ | + \rangle = 1$
 $\langle - | P_- | + \rangle = 0.$

Second SG magnet does nothing.

b) $SG \hat{z}(+) \rightarrow SG \hat{x} \rightarrow \text{measure}$
 $p_+ = |\langle S_x; + | S_z; + \rangle|^2 = \cos^2(\frac{\pi}{4}) = 1/2$
 $p_- = |\langle S_x; - | S_z; + \rangle|^2 = \sin^2(\frac{\pi}{4}) = 1/2$
 50/50 split

Definite $\hat{S}_z$ has largest $\hat{S}_x$ spread.

c) $SG \hat{z}(+) \rightarrow SG \hat{x}(+) \rightarrow SG \hat{z}$

Filters act in sequence

$$P_- = |\langle S_z; - | S_x; + \rangle|^2 |\langle S_x; + | S_z; + \rangle|^2 = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

This is the physics behind
 $[\hat{S}_x; \hat{S}_z] \neq 0$ and $\Delta S_x \Delta S_z = \hbar/4$

d) Some general angle
 $SG \hat{z}(+) \rightarrow \underbrace{SG \hat{n}(+)}_{\beta \text{ angle}} \rightarrow SG \hat{z}$

$$p_- = \cos^2\left(\frac{\beta}{2}\right) \sin^2\left(\frac{\beta}{2}\right) = \frac{1}{4} \sin^2 \beta$$

$$p_+ = \cos^4\left(\frac{\beta}{2}\right)$$

n.b. $p_+ + p_- = \cos^2(\beta/2) \leq 1.$

$\beta = 0 \rightarrow \hat{z} \hat{z} \hat{z}$ nothing happens

$\beta = \pi \rightarrow \hat{z} - \hat{z} \hat{z}$ nothing goes through

$\beta = \pi/2 \rightarrow \hat{z} \hat{x} \hat{z} \quad |+\rangle \rightarrow 50/50 \rightarrow 25/25$